seconds, during which time the particles make many millions of orbits. Each burst of particles may contain $\sim 10^{12}$ particles, and in the Berkeley machine there was one burst every 6 s. The successful operation of the synchrotron depends on the existence of phase stability.

The Bevatron (later the Bevalac) operated from 1954 to 1993. Larger and more recent (proton) accelerators are the 76 GeV machine of Serpukhov (USSR), and the 500 GeV machine at Batavia in the USA. The Batavia machine contains 1000 magnets along its 6.3 km orbit, each magnet weighing 10 tons.

At CERN, a machine was built that could produce 10^{13} 400 GeV protons s^{-1}; the acceleration diameter was 2.2 km. The injection time (with 10 GeV protons) was 23 s (to fill the acceleration ring), while the acceleration time was 9 s; for each cycle 2.5 MeV was added in energy. The field strength of the 200 bending magnets was 1.8 T and the power consumption 50 MW.

Still higher energies have been achieved in collisions between particles circulating in two beam trains in opposite direction by using intersecting storage rings (ISR). In this system the energy loss due to momentum conservation in the reaction between high speed projectiles and stationary target atoms is eliminated. At CERN proton collision energies of 7 TeV have been obtained and the system is being upgraded to 8 TeV in 2012.

Synchrotrons have been built in which electrons are accelerated in a fixed orbit in a strong magnetic field ($\geq$1 T) to energies in the GeV range. The intense electromagnetic (synchrotron) radiation has usually a maximum intensity at $\leq$1 keV and is used for studies of photon induced reactions (e.g. for meson production), for radiation chemistry, as a strong light source, etc.

A special type of synchrotron is the so called storage ring. In a storage ring the particles are kept at a constant velocity generating a constant flux of syncrotron radiation. In order to maximise the synchrotron radiation intensity these rings are often equipped with wrigglers or undulators which give the beam multiple direction changes, see §16.9.1.

16.8. NEUTRON GENERATORS

Many radioactive isotopes are most conveniently produced by irradiation of a target with neutrons. There are three main ways of obtaining useful neutron fluxes:

(a) through accelerator-induced reactions (Table 16.1);
(b) through nuclear reactors (Ch. 19);
(c) through nuclear reactions induced by radioactive isotopes (§10.9).

16.8.1. Reactions

Table 16.1 gives the most common accelerator-induced nuclear reactions leading to the production of neutrons. In general, the neutron yield increases with increasing beam energy up to some maximum value of the latter after which competition of other nuclear

Table 16.1 Neutron yield for some accelerator produced reactions (see also Fig. 16.10)

Reaction	*Q*(MeV)	Target	Yield(*) (n s^{-1} μA^{-1})	
$^{7}Li(p,n)^{7}Be$	−1.64	Li salt	10^{7}	for 2.3 MeV H^{+}
$^{2}D(d,n)^{3}He$	3.27	D_2O ice	7×10^{7}	for 1 MeV D^{+}
$^{3}H(d,n)^{4}He$	17.6	^{3}H in Ti	2×10^{7}	for 0.1 MeV D^{+}
$^{9}Be(d,n)^{10}B$	3.76	Be metal	2×10^{10}	for 7 MeV D^{+}
$^{9}Be(\gamma,n)^{8}Be$	−1.67	Be metal	10^{9}	for 10 MeV e
$U(\gamma,fp)$	−5.1	U metal	10^{11}	for 40 MeV e

(*) 1 $\mu A = 6.24 \times 10^{12}$ single-charged ions per second.

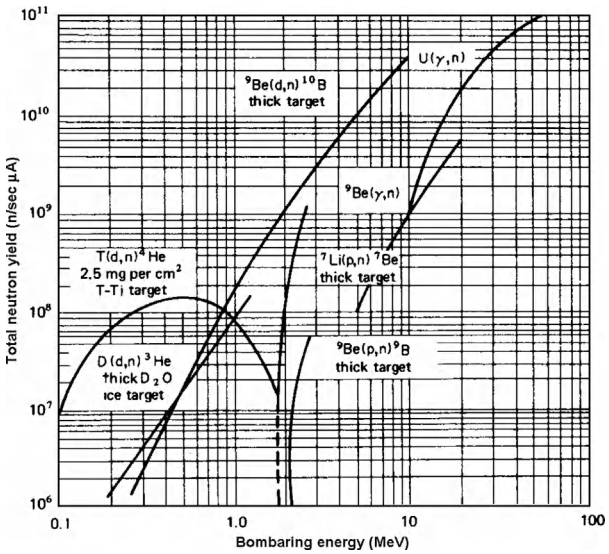


Figure 16.11 Neutron yield as function of projectile energy.

reactions reduces the probability of the initial neutron producing reaction (Fig. 16.11). The least expensive way to get a neutron flux of sufficient intensity for producing radioactive nuclides in easily measurable quantities (e.g. for activation analysis) is through the T + D reaction using a Cockcroft-Walton accelerator. The reaction

$$^{7}Li + {}^{1}H \rightarrow {}^{7}Be + {}^{1}n \tag{16.13}$$

is used for the production of mono-energetic neutrons (cf. §11.3). Higher neutron yields can be obtained by the reaction of ^{9}Be with deuterons. Neutrons can also be obtained from ^{9}Be by irradiation in electron accelerators. The passage of the electrons in the target results in the production of high energy Bremsstrahlung γ-rays (§7.4.2) which react with the beryllium to produce neutrons. An even higher neutron flux is obtained for the same reason when uranium is bombarded by electrons.

Extremely intense neutron sources can be constructed from a high energy ($\sim$1 GeV) high current proton linear accelerator and a heavy element (e.g. molten Pb) spallation target. Tentative designs have been described that would be capable of generating $\sim$0.1 g of neutrons (0.1 mole) per second. Such intense neutron sources could be used to transmute longlived radioactive nuclides (into inactive or shorter lived ones) or drive a subcritical assembly for production of energy by fission of actinides.

16.9. AREAS OF APPLICATION FOR ACCELERATORS

Let us consider a few uses of accelerators. In §16.8 we have discussed the use of accelerators for neutron production. Electron as well as positive particle beams from these accelerators can be used directly for the study of radiation effects in materials (see Ch. 8), to produce X-rays, or to generate intense light. Accelerators are used in medicine for producing intense γ- or X-ray beams for cancer treatment, or charged particle beams for similar purposes (the "proton-knife"). Accelerators in the 10 MeV range are very useful in producing radionuclides, which either cannot be made through (n,γ)-reactions in nuclear reactors or are very shortlived. Small cyclotrons of a size suitable for use in hospitals are commercially available. The precise definition of energy available in van de Graaff accelerators makes them particularly useful for the study of the energy levels of nucleons in nuclides and for ^{14}C dating. VdG's, cyclotrons, linear accelerators, and synchrotrons have been extensively used for the study of nuclear reactions.

Very high energy accelerators are used for the production and study of elementary particles. Once the beam energy exceeds a couple of hundred MeV, mesons and other elementary particles are produced in nuclear reactions and can be studied directly or they can be used to cause various types of nuclear reactions.

16.9.1. Syncrotron light sources

High energy electron accelerators, mostly in form of large diameter storage rings, are used to produce high intensity synchrotron radiation for basic studies in chemistry, material science and biology. The synchrotron "light" is usually obtained by placing bending magnets in the path of the circulating electron beam. A series of opposing strong magnets repeatedly bend the beam back and forth many times. These devices are characterized by their K_U-value which is defined as given below.

$$K_U = \frac{eB\lambda}{2\pi v m_e} \tag{16.14}$$

where v is the electron velocity in the beam and λ is the distance beween magnets with the same orientation. When K_U is $\ll 1$ the device is called an undulator and produces x-rays in a narrow energy band, but when K_U is $\gg 1$ the device is called a wiggler and produces a continuum of x-rays within a rather wide energy range. A typical device is